

```
In [2]: # Scraping Pokemon data from pokemon database and saving to CSV
import requests
from bs4 import BeautifulSoup
import pandas as pd

url = 'https://pokeмонdb.net/pokedex/all'
response = requests.get(url)
soup = BeautifulSoup(response.text, 'html.parser')

# Find the main standings table
table = soup.find('table')
df = pd.read_html(str(table))[0] # Convert HTML table to pandas DataFrame

# df.to_csv('pokedex.csv', index=False)

C:\Users\yashb\AppData\Local\Temp\ipykernel_25836\4288941770.py:12: FutureWarning: Passing literal html to 'read_html' is deprecated and will be removed in a future version. To read from a literal string, wrap it in a 'StringIO' object.
df = pd.read_html(str(table))[0] # Convert HTML table to pandas DataFrame
```

```
In [3]: df
```

	#	Name	Type	Total	HP	Attack	Defense	Sp. Atk	Sp. Def	Speed
0	1	Bulbasaur	Grass Poison	318	45	49	49	65	65	45
1	2	Ivysaur	Grass Poison	405	60	62	63	80	80	60
2	3	Venusaur	Grass Poison	525	80	82	83	100	100	80
3	3	Venusaur Mega Venusaur	Grass Poison	625	80	100	123	122	120	80
4	4	Charmander	Fire	309	39	52	43	60	50	65
...	...	...	...	...	...	...	...	...	...	...
1210	1023	Iron Crown	Steel Psychic	590	90	72	100	122	108	98
1211	1024	Terapagos Normal Form	Normal	450	90	65	85	65	85	60
1212	1024	Terapagos Terastal Form	Normal	600	95	95	110	105	110	85
1213	1024	Terapagos Stellar Form	Normal	700	160	105	110	130	110	85
1214	1025	Pecharunt	Poison Ghost	600	88	88	160	88	88	88

1215 rows × 10 columns

```
In [13]: # splitting 'Type' column into 'Type1' and 'Type2'
types_split = df['Type'].str.split(' ', expand=True)
df['Type1'] = types_split[0]
df['Type2'] = types_split[1]
```

```
In [14]: # checking for overall data quality
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1215 entries, 0 to 1214
Data columns (total 12 columns):
# Column Non-Null Count Dtype
---
0 # 1215 non-null int64
1 Name 1215 non-null object
2 Type 1215 non-null object
3 Total 1215 non-null int64
4 HP 1215 non-null int64
5 Attack 1215 non-null int64
6 Defense 1215 non-null int64
7 Sp. Atk 1215 non-null int64
8 Sp. Def 1215 non-null int64
9 Speed 1215 non-null int64
10 Type1 1215 non-null object
11 Type2 669 non-null object
dtypes: int64(8), object(4)
memory usage: 114.0+ KB
```

```
In [15]: # checking for missing values
df.isnull().sum()
```

```
Out[15]: #
Name 0
Type 0
Total 0
HP 0
Attack 0
Defense 0
Sp. Atk 0
Sp. Def 0
Speed 0
Type1 0
Type2 346
dtype: int64
```

```
In [16]: # Visualizing Pokémon data
df.describe()
```

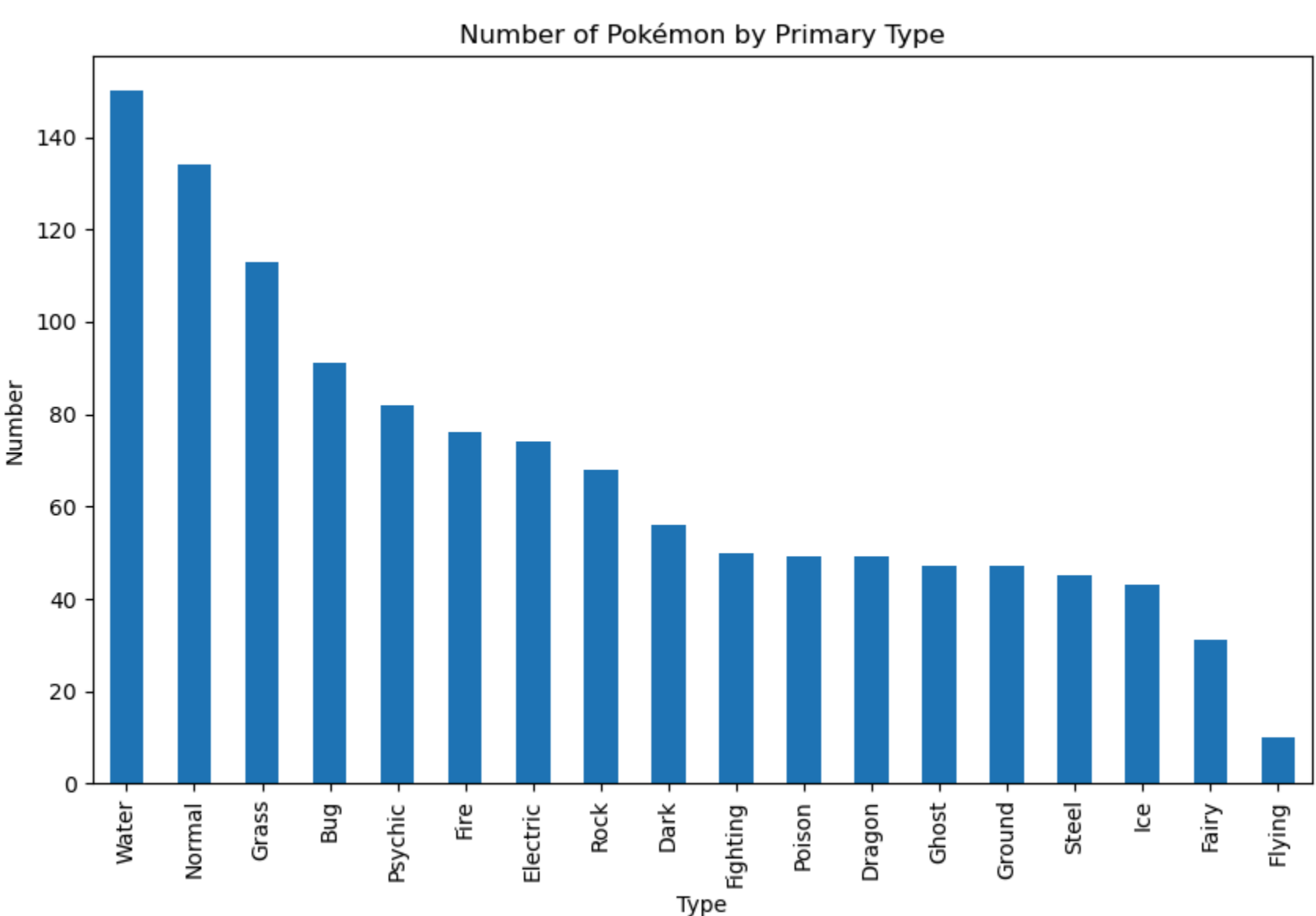
```
Out[16]:
```

	#	Total	HP	Attack	Defense	Sp. Atk	Sp. Def	Speed
count	1215.000000	1215.000000	1215.000000	1215.000000	1215.000000	1215.000000	1215.000000	1215.000000
mean	501.735802	443.104527	71.244444	81.152263	75.007407	73.224691	72.441152	70.034568
std	298.979081	121.193406	26.927819	32.037134	30.740999	32.757152	27.578188	30.161298
min	1.000000	175.000000	1.000000	5.000000	5.000000	10.000000	20.000000	5.000000
25%	240.500000	332.000000	52.000000	57.000000	52.000000	50.000000	51.000000	45.000000
50%	495.000000	465.000000	70.000000	80.000000	70.000000	65.000000	70.000000	68.000000
75%	753.500000	521.000000	85.000000	100.000000	91.000000	95.000000	90.000000	91.000000
max	1025.000000	1125.000000	255.000000	190.000000	250.000000	194.000000	250.000000	200.000000

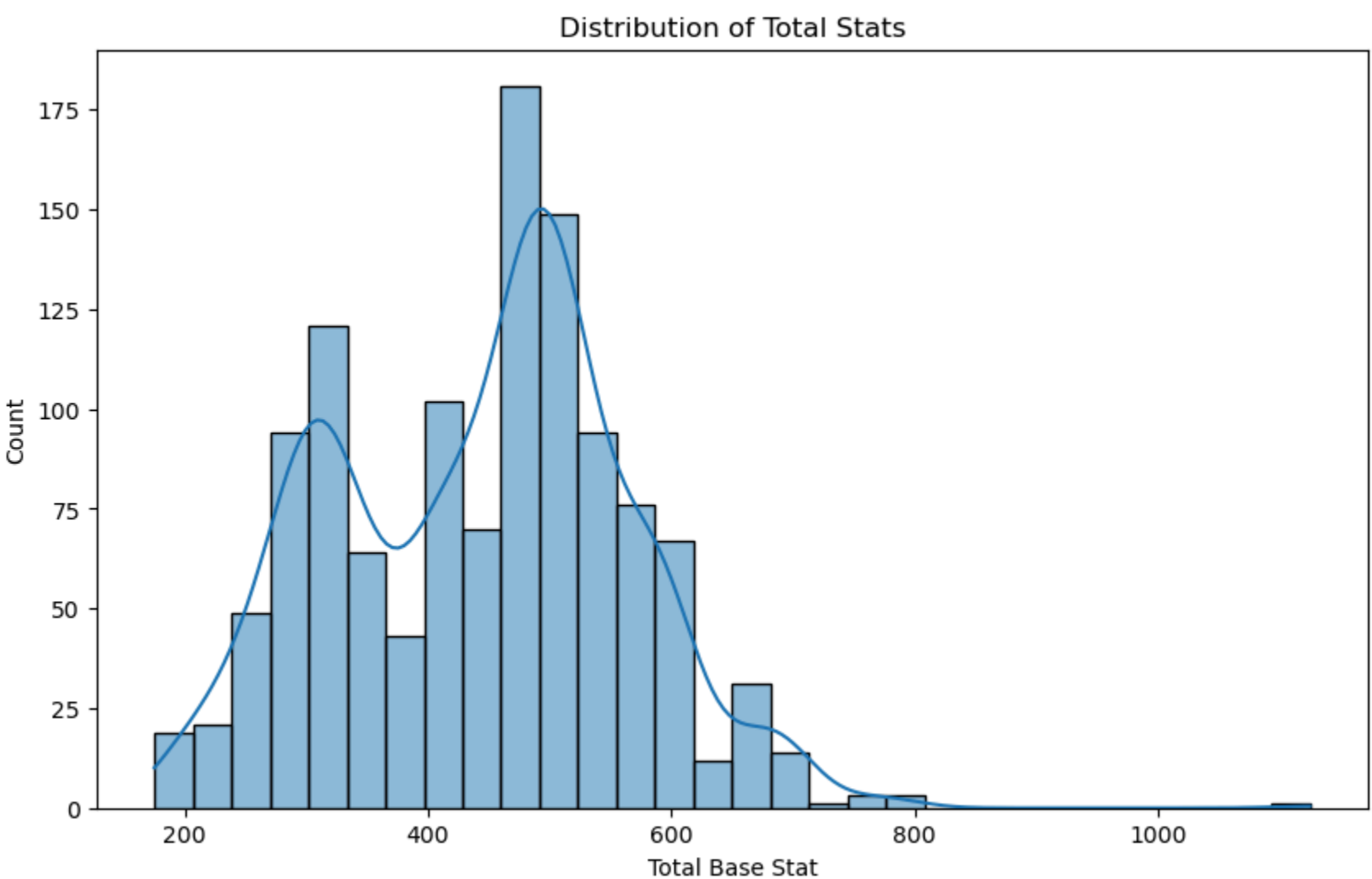
```
In [17]: df['Type1'].value_counts()
```

```
Out[17]: Type1
Water 150
Normal 134
Grass 113
Bug 91
Psychic 82
Fire 76
Electric 74
Rock 68
Dark 56
Fighting 50
Poison 49
Dragon 49
Ghost 47
Ground 47
Steel 45
Ice 43
Fairy 31
Flying 10
Name: count, dtype: int64
```

```
In [18]: # visualizing the primary type distribution
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(10,6))
df['Type1'].value_counts().plot(kind='bar')
plt.title('Number of Pokémon by Primary Type')
plt.xlabel('Type')
plt.ylabel('Number')
plt.show()
```

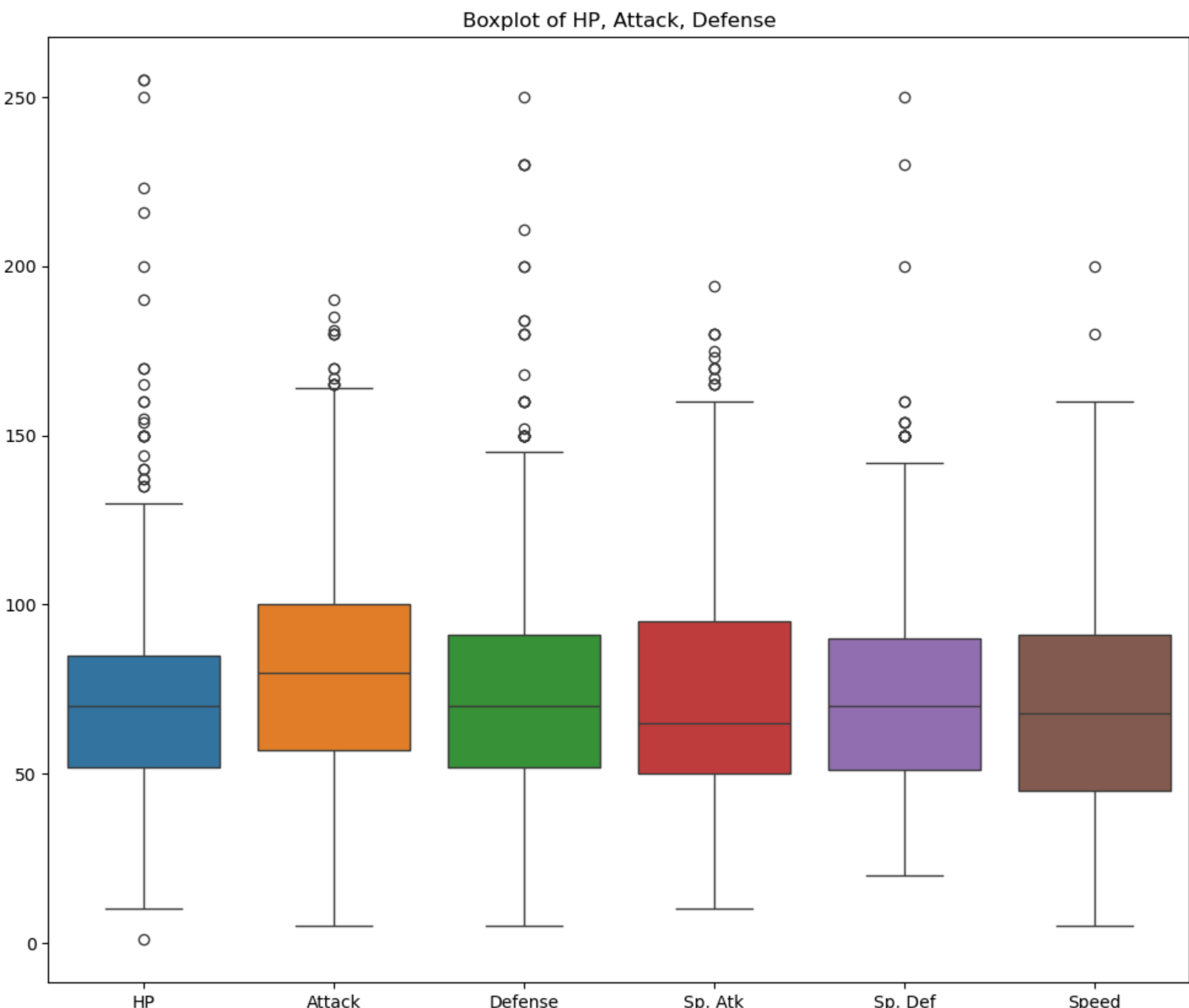


```
In [19]: # visualizing the distribution of base stats of pokemon
plt.figure(figsize=(10,6))
sns.histplot(df['Total'], bins=30, kde=True)
plt.title('Distribution of Total Stats')
plt.xlabel('Total Base Stat')
plt.ylabel('Count')
plt.show()
```



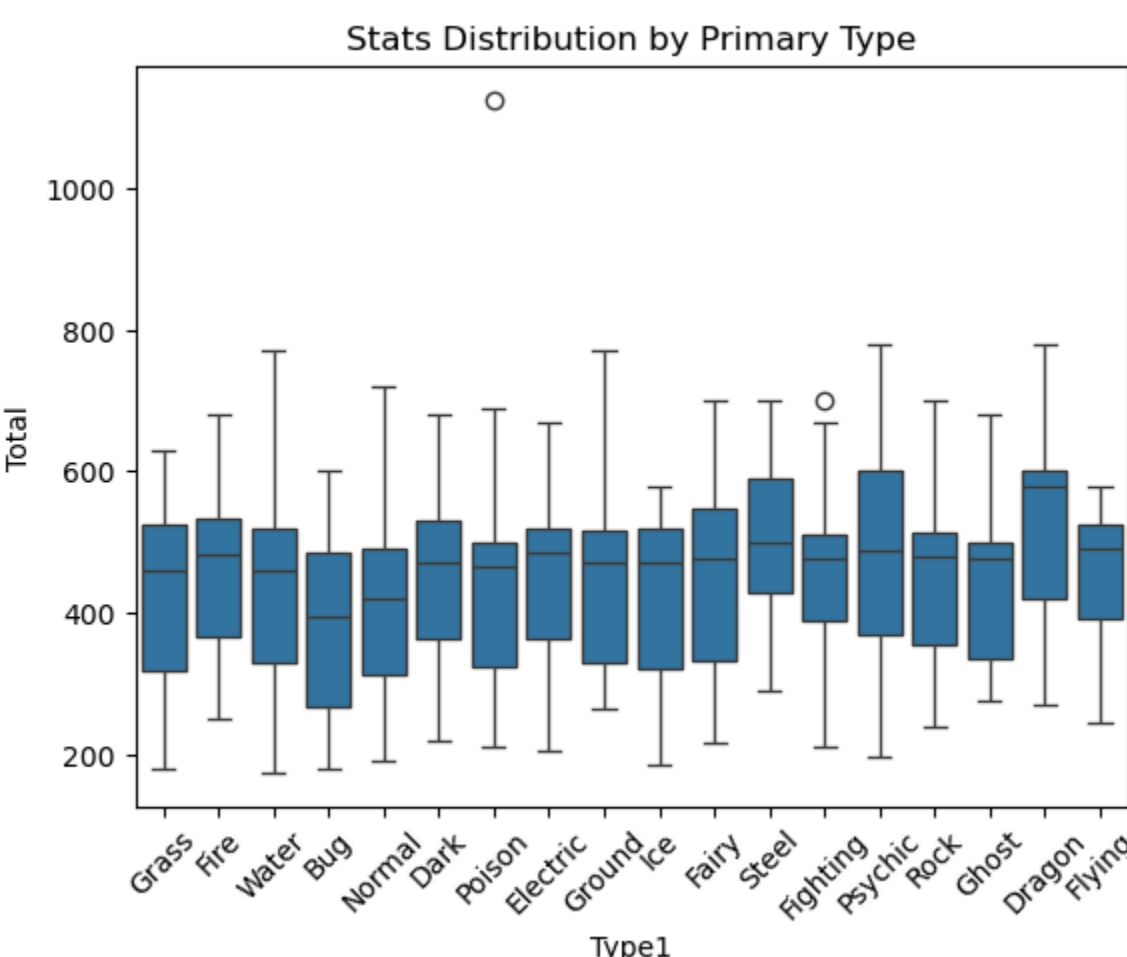
As We can see in the boxplot below there are very less outliers for speed as compared to other stats because in the game speed is important but not directly useful hence not many extreme pokemons for speed. The same can be said about Sp.Def. stats as most players prefer attack based styles.

```
In [20]: # visualizing boxplot of all stats
stats = ['HP', 'Attack', 'Defense', 'Sp. Atk', 'Sp. Def', 'Speed']
plt.figure(figsize=(12,10))
sns.boxplot(data=df[stats])
plt.title('Boxplot of HP, Attack, Defense')
plt.show()
```



The boxplot below is showing the stats distribution by primary type. Nearly all types have a similar mean except for the dragon type as Dragons are considered as legendary creatures and hence most of the Dragon types in the game are legendary hence their stats are skewed at a larger value as compared to other types.

```
In [21]: sns.boxplot(data=df, x='Type1', y='Total')
plt.title('Stats Distribution by Primary Type')
plt.xticks(rotation=45)
plt.show()
```




```
In [23]: df = df.drop(columns=['Type', 'Type2']) # Drop original Type column and Type2 as they are no longer needed
df = df.rename(columns={'Type1': 'Primary Type'}) # Rename Type1 to
df.to_csv('pokedex_cleaned.csv', index=False) # Save cleaned DataFrame to CSV
```

In this Notebook I Scraped data about pokemon a famous franchise and used various plots to show the distribution of various stats in between pokemon of various types and also showed the differences between the stats and also the variety of pokemon according to their total stats.